

Liver Failure

Hepatic Encephalopathy

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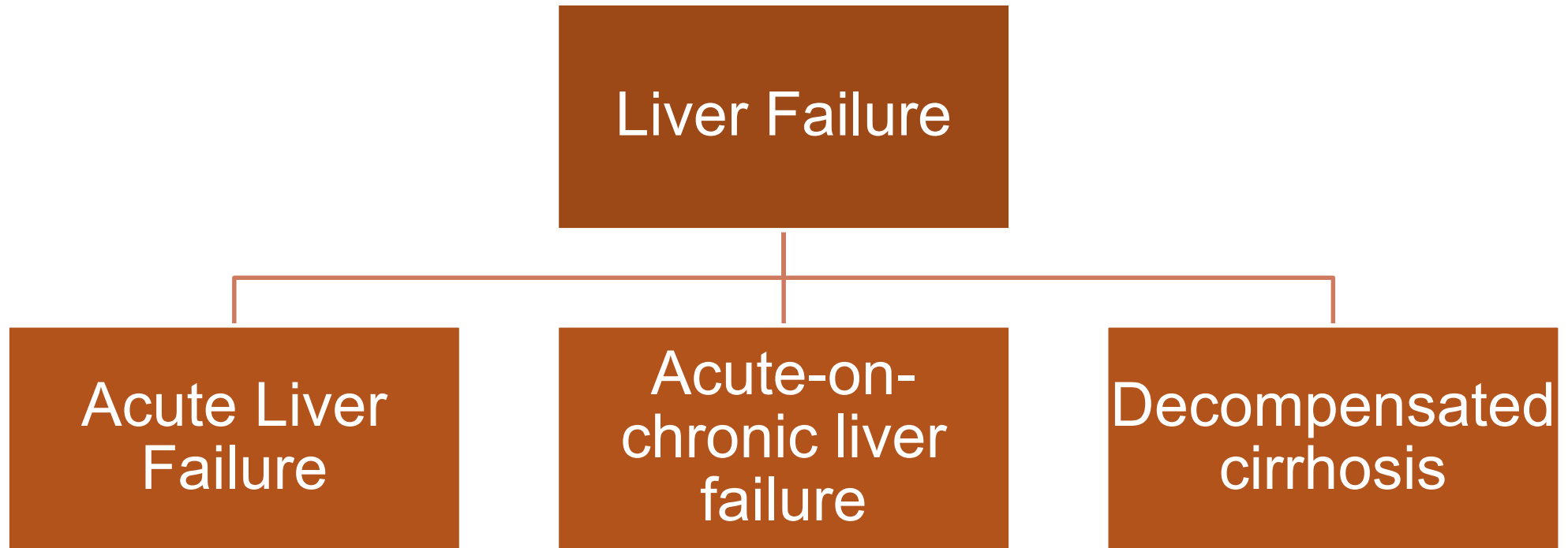
香港大學深圳醫院
The University of Hong Kong - Shenzhen Hospital



WCS Outline

- Liver failure: definitions and categorization
- Etiologies
- Prognostic factors
- Hepatic encephalopathy
 - Precipitating factors, diagnosis, management
- Management of liver failure

Liver Failure: Categorization



Acute Liver Failure

- Development of severe acute liver injury with encephalopathy and impaired synthetic function (INR >1.5) in a patient **without pre-existing** cirrhosis / chronic liver disease

Acute Liver Failure - Causes

- Drug-related
 - Paracetamol overdose
 - Idiosyncratic drug reactions
 - Herbal-related
- Acute viral hepatitis (HAV / HEV / acute HBV)
- Pregnancy-related causes (acute fatty liver of pregnancy / HELLP syndrome) - rare
- Other causes also rare

Clinical Features of ALF: Not obvious

Whole body

- Systemic inflammatory response^{1,2}
- High energy expenditure and catabolism^{3,4}

Liver

- Loss of metabolic function
- Decreased gluconeogenesis leading to hypoglycaemia
- Decreased lactate clearance leading to lactic acidosis
- Decreased ammonia clearance leading to hyperammonaemia
- Decreased synthetic capacity leading to coagulopathy

Lungs

- Acute lung injury
- Adult respiratory distress syndrome

Adrenal gland

- Inadequate glucocorticoid production contributing to hypotension⁵

Bone marrow

- Frequent suppression, especially in viral and seronegative disease⁶

Circulating leucocytes

- Impaired function and immunoparesis contributing to high risk of sepsis⁷

Brain

- Hepatic encephalopathy
- Cerebral oedema
- Intracranial hypertension⁸

Heart

- High output state
- Frequent subclinical myocardial injury⁹

Pancreatitis

- Particularly in paracetamol-related acute liver failure¹⁰

Kidney

- Frequent dysfunction or failure²

Portal hypertension

- Might be prominent in subacute disease and confused with chronic liver disease

Acute-on-chronic liver failure

- Acute liver insult manifesting jaundice and INR >1.5 , complicating within 4 weeks by ascites and / or encephalopathy in patients **with underlying chronic liver disease**
- Chronic liver disease **might be undiagnosed**
- Increased mortality (28-day: $>20\%$)

Acute-on-chronic liver failure: examples

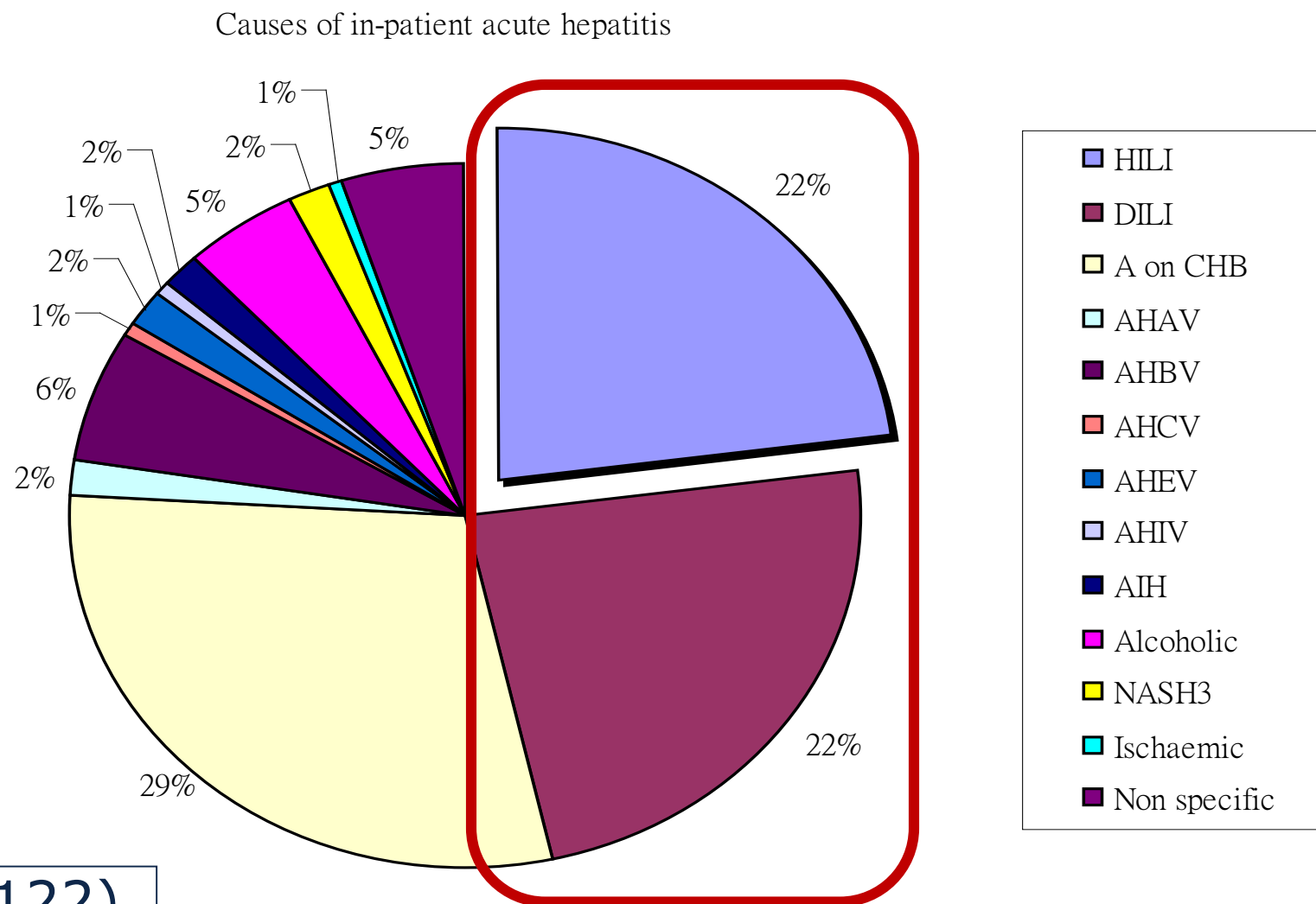
- HBV-related:
 - Severe exacerbation of HBV
 - Immunosuppressive agents (steroids, rituximab etc)
- Infections:
 - Superimposed HAV / HEV
 - Other systemic infections (Pneumonia / UTI / soft tissue etc)
- External agents
 - Alcoholic hepatitis
 - Hepatotoxic drugs / herbs

Specific “Hepatotoxic” Drugs

- Paracetamol (overdose)
- NSAIDs
- Statins (actually very rare)
- Anti-TB (isoniazid, rifampicin, pyrazinamide)
- Other Anti-microbials
- Anticonvulsants
- Rheumatological (methotrexate, azathioprine)
- Amiodarone

Usually Idiosyncratic

Role of Herb-related Hepatotoxicity in an Acute HA Hospital [>10 years ago]



102 types of hepatotoxic herbs reported in literature review

Chinese name	Latin name
丁香	Flos Caryophylli
土茯苓	Rhizoma smilacis Glabrae
大黃	Radix et Rhizoma Rhei
川楝子	Melia toosendan
火麻仁	Fructus cannabis
半夏	Rhizoma Pinelliae praeparatum
白芍	Radix Paeonie Alba
白花丹	Herba Plumbaginis Zeylanicae
白鮮皮	Cortex Dictamni
石榴皮	Pericarpium Granati
石蠟香科	Teucrium chamaedrys
合歡皮	Cortex Albiziae
艾葉	Artemisia argyi Levl. Eet Vant
何首烏	Radix Polygoni Multiflori
決明子	Cassia obtusifolia
牡丹皮	cortex Moutan
赤芍	Radix Paeoniae Rubra
防己	Radix Stephaniae Tetrandrae
延胡索	Rhizoma corydalis
虎杖	Rhizoma et Radix polygoni cuspidati
金果欖	Radix Tinosporae
青黛	Lapis Chloriti
苦杏仁	Semen Armeniacae Amarum

Chinese name	Latin name
重樓	Rhizoma paridis
柴胡	Radix Bupleuri
茯苓	Poria
梔子	Fructus Gardeniae
淫羊藿	Herba Epimedii
細辛	Radix et Rhizoma Asari
魚膽	fish gall bladder
麻黃	Herba Ephedrae
莢術	Rhizoma Zedoariae
番瀉葉	Cassia acutifolia and Cassia angustifolia
紫苑	Radix et Rhizoma Asteris
黃藥子	Dioscorea bulbifera
蜈蚣	Scolopendra subspinipes
補骨脂	Fructus Psoraleae
蒲黃	pollen typhae
蒼耳子	Fructus Xanthii
豬膽	pig gall bladder
澤瀉	Alisma orientale
獨活	Radix Angelicae Pubescentis
蘇鐵	Folium cycadis
靈芝	Ganoderma lucidum
鬱金	Radix Curcumae

清水
碗
煎至
碗
分
温服

白芍	18.75 (g)
金銀花	11.25 (g)
秦艽	11.25 (g)
澤蘭	11.25 (g)
熟地黃	15 (g)
生地黃	15 (g)
補骨脂	11.25 (g)
杜仲	11.25 (g)
黨參	18.75 (g)
黃芩	11.25 (g)
柴胡	15 (g)
法半夏	18.75 (g)
蔓荊子	11.25 (g)
皂角刺	18.75 (g)
地骨皮	18.75 (g)
桔梗	11.25 (g)

TCM in tablet / capsule form



誤食毒菇 教師暫毋須換肝

80,248



A A

■誤食毒菇的胡東兵（左）情況好轉，其兒子胡君儒（右）可以鬆一口氣。梁志永攝



【本報訊】因誤食有毒小托柄鵝膏菌致急性肝衰竭的中學教師胡東兵和妻子，昨天均獲好消息，胡東兵經搶救後開始「解毒」，暫不需要換肝，胡太則獲捐屍肝，昨晚10時完成換肝手術。胡東兵昨在深切治療部接受訪問時稱，絕不會再胡亂採摘野菇。



妻子獲屍肝移植

胡氏夫婦日前行山時因採摘了有毒的小托柄鵝膏菌回家食用，雙雙中毒致急性肝衰竭，情況危殆，其任職助理工程師的兒子胡君儒，在父母出事後四出呼籲有心人捐肝。病情較危急的胡太，昨晨進行屍肝移植手術，胡東兵則繼續留瑪麗醫院深切治療部。瑪麗醫院肝膽外科副顧問醫生馮恩裕昨天稱，胡東兵對治療肝衰竭的藥物

Mushroom poisoning causing acute liver failure

57歲女子富亨邨摘野菇進食後疑中毒 嘔吐、腹瀉入院



富亨邨附近採食野菇 疑中毒 大埔中年婦嘔吐腹瀉入院

社會新聞

讚好 15

撰文：李恩慈 2020-05-29 16:02 最後更新日期：2020-05-29 16:54



衛生署衛生防護中心今日（29日）公布一宗懷疑進食野生菇類後引致食物中毒的個案。一名57歲女子昨日於大埔富亨邨附近採摘野生菇類，今日在家中進食約兩小時後，出現腹痛、嘔心、嘔吐和腹瀉等徵狀。病人到大埔那打素醫院急症室求診，目前情況穩定。

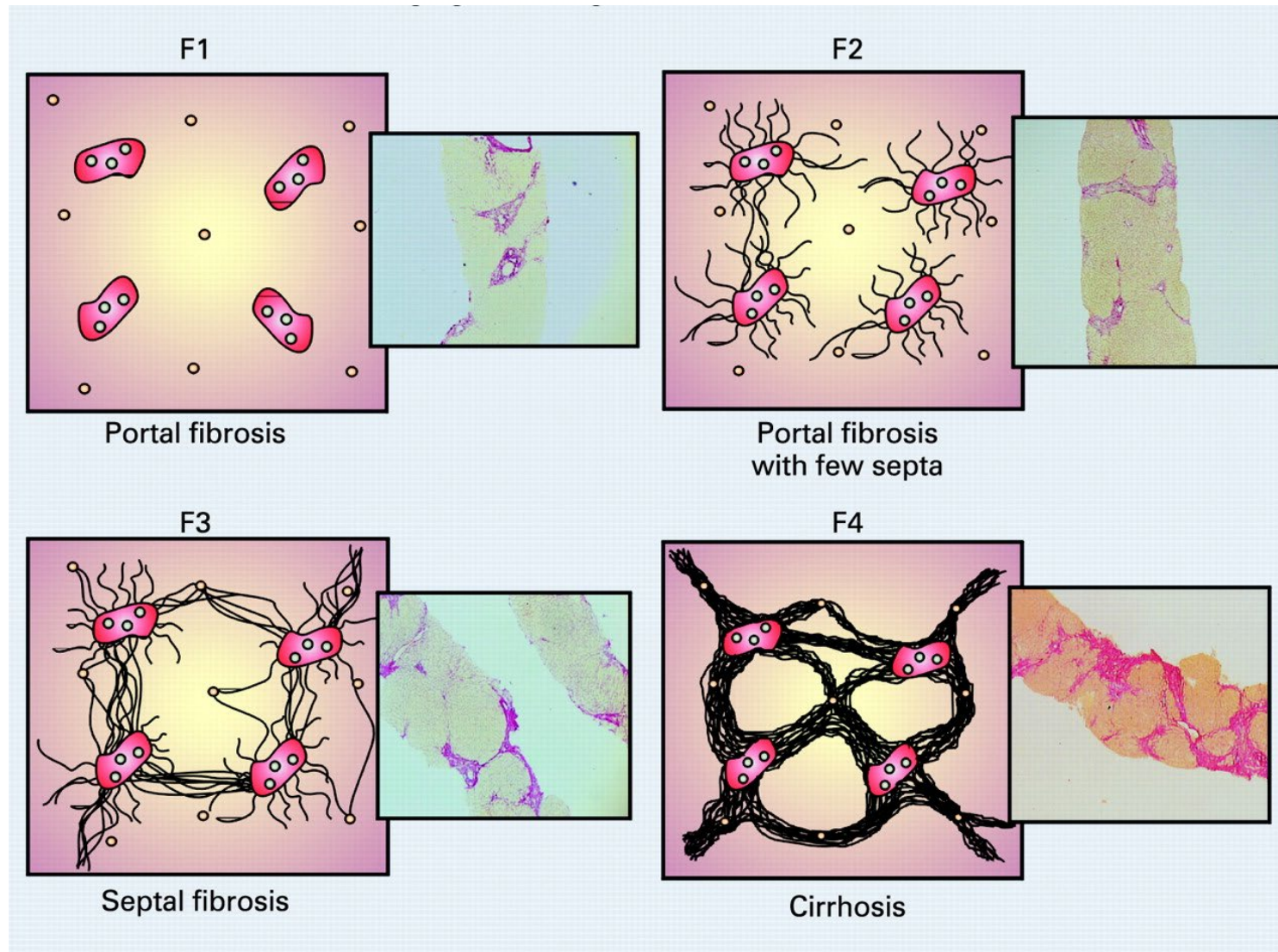
衛生防護中心發言人表示，進食菇類引致的食物中毒一般為急性，病徵包括短時間內出現嘔心、嘔吐和腹痛等腸道徵狀。因應不同的菇種，病人亦可能出現如大量流汗、幻覺、昏迷或其他神經系統病徵和肝衰竭。中心呼籲市民切勿在公園或郊野採摘野生菇類進食。

Systemic Effects of Hepatocellular Failure

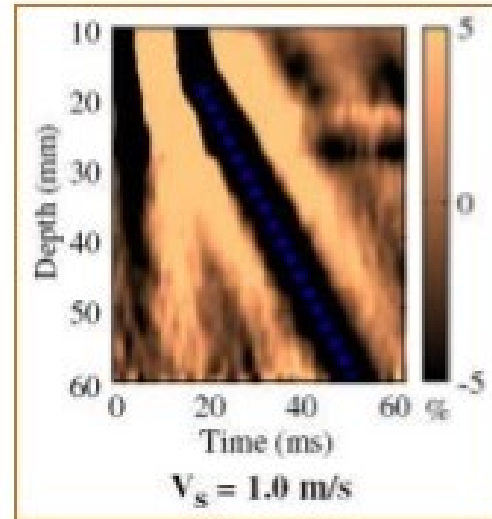
- General non-specific malaise
- Encephalopathy
- Ascites
- Jaundice (usually late stage)
- Fetor hepaticus (usually late stage)
- Circulatory changes:
 - ↑ CO, ↓ peripheral resistance,
 - ↓ renal blood flow
- N2 metabolism:
 - ↑ NH₃, ↓ urea, amino acid abnormalities,
 - ↓ albumin
- Skin changes due to vasodilatation:
 - spider angioma,
 - palmer erythema
- Endocrine changes:
 - hypogonadism, hyperaldosteronism
- Defective coagulation, ↓ platelet, ↑ fibrinolysis

There may not be
concomitant hepatitis

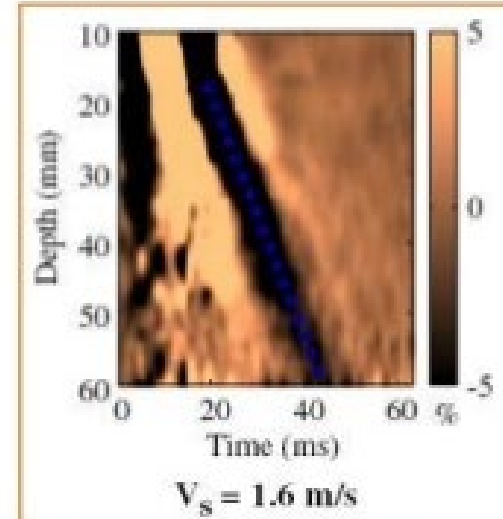
Fibrosis and Cirrhosis



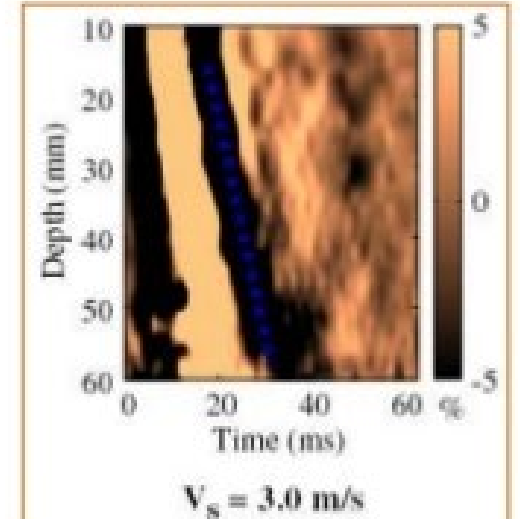
Vibration controlled transient elastography – the non-invasive method to stage fibrosis



$E = 3.0 \text{ kPa}$
F 0



$E = 7.7 \text{ kPa}$
F 2



$E = 27 \text{ kPa}$
F 4

Prognostic Factors for Cirrhosis

Child-Pugh Score

	1 point	2 points	3 points
Bilirubin (umol/L)	<34	34-51	>51
Albumin (g/L)	>35	28-35	<28
INR	<1.7	1.7-2.3	>2.3
Ascites	None	Mild	Refractory
Encephalopathy	None	Gr 1-2	Gr 3-4
Child classes:	A	5 - 6 points	
	B	7 - 9 points	
	C	10 - 15 points	

* For bilirubin, 1 mg/dl = 17.1 μ mol/L

Prognostic Factors for Cirrhosis

Child-Pugh Score

Disadvantages:

- limited discrimination
 - only 8 levels between Child B and C
 - same score for different values of bilirubin, albumin, PT, etc.
- Subjective assessment of ascites and encephalopathy
- Variability of PT and albumin in different laboratories
- Albumin and PT measurements can be “manipulated”

Prognostic Factors for Cirrhosis

MELD score

- **Model for End-stage Liver Disease (MELD)**

score from Mayo Clinic, USA

$$\begin{aligned} &= 3.8 \times \log_e (\text{bilirubin mg/dL}) + \\ &\quad 11.2 \times \log_e (\text{INR}) + \\ &\quad 9.6 \times \log_e (\text{creatinine mg/dL}) + 6.4 \end{aligned}$$

Very good in prediction of 3-month mortality, useful to
prioritize patients for liver transplant

Prognostic Factors for Cirrhosis

MELD score v 2.0

- **Model for End-stage Liver Disease (MELD)-Na** score from Mayo Clinic, USA
(2016 onward)

Very good in prediction of 3-month mortality, useful to prioritize patients for liver transplant

MELDNa/MELD-Na Score for Liver Cirrhosis



Adds sodium to the MELD model for liver cirrhosis.

INSTRUCTIONS

Note: The Organ Procurement and Transplantation Network recently incorporated sodium into their [new MELD Score](#) as of January 2016, using a different model than MELD-Na. We recommend using the [new MELD Score](#) primarily.

Why Use ▼

Dialysis at least twice in the past week

No

Yes

Creatinine

Norm: 62 - 115

μmol/L ↔

Bilirubin

Norm: 5.13 - 32.49

μmol/L ↔

INR

Norm: 0.8 - 1.2

Sodium

Norm: 136 - 145

mmol/L ↔

Poor Prognostic Factors for Acute Liver Failure

– King's Criteria

- Irrespective of grade of encephalopathy
 - PT > 100 sec
- or 3 of the following*
 - age < 10 or > 40
 - duration of jaundice before encephalopathy > 7 days
 - PT > 50 sec
 - bilirubin > 300 $\mu\text{mol/l}$
 - non-A non-B, halothane or idiosyncratic drug reaction

Poor Prognostic Factors for Acute Liver Failure

- For paracetamol poisoning
 - pH < 7.3
 - or* - PT > 100 sec and creatinine > 300 $\mu\text{mol/l}$ in grade III or IV encephalopathy

Prognostic Factors for ACLF

- Depends on 6 factors:
 - Cerebral: **HE**
 - Respiration: **SaO₂** / FiO₂
 - Circulation: Need of **vasopressor**
 - Liver: **Bilirubin** level
 - Coagulation: **INR** level
 - Kidney: **Creatinine** level
- MELD score still useful

Associated Complications of Liver Failure

- Infections
- Variceal bleeding
- Ascites / Spontaneous bacterial peritonitis
- Hepatorenal syndrome
- Hepatic encephalopathy
- (Coagulopathy)

Infections in Liver Failure

- Reticuloendothelial dysfunction and reduced opsonization
- Very common (espc respiratory/urinary)
- Bacteremia up to 25% in fulminant hepatic failure
- Staph, strep and GNR most common
- Broad spectrum antibiotic prophylaxis recommended
- Risk of fungal infection – esp. Candida

Hepatic Encephalopathy in decompensated cirrhosis

Causes

- Cirrhosis
 - spontaneous portal-systemic shunts
 - progressive hepatic failure in detoxification
- Additional precipitating factors for cirrhotic patients
 - $\uparrow$ N₂ products thru diet
 - sepsis
 - GI bleed
 - renal failure
 - constipation

Hepatic Encephalopathy in decompensated cirrhosis

Causes

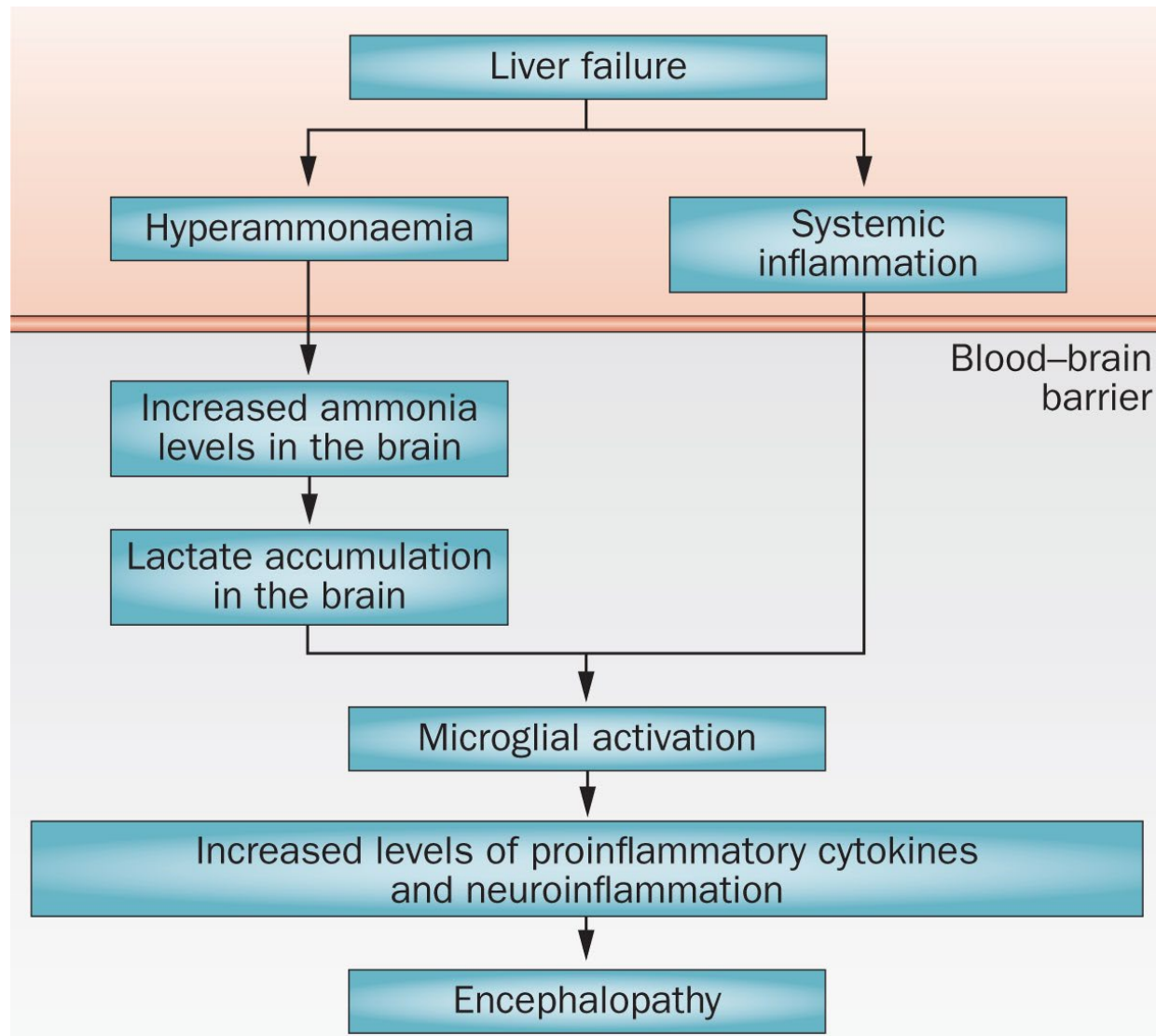
- Additional precipitating factors for cirrhotic patients
 - ↓ vascular vol → ↓ O₂ to liver, e.g.,
 - GI bleed
 - overdiuresis
 - excess paracentesis with inadequate albumin replacement
 - diarrhoea and vomiting
 - Sedatives
 - Electrolyte imbalance esp. ↓ K
 - Acid base imbalance
 - Artificial P-S shunts
 - TIPS
 - Surgery

Hepatic Encephalopathy

Causes

- Fulminant (Acute / ACLF)
 - viral hepatitis
 - drugs, e.g., paracetamol, halothane, INAH
 - metabolic, e.g., pregnancy, Reye syndrome, Wilson's disease
 - CVS, e.g., shock, heat stroke
 - herbal medicines and health food products

Hepatic Encephalopathy: Underlying Mechanism



Hepatic Encephalopathy

Pathogenesis

(I) ↑ arterial ammonia

- ↓ ability of liver to detoxify NH_3 to urea
- ↑ cerebral metabolic rate and permeability to NH_3
- ↑ $\text{NH}_3 \rightarrow$ astrocyte damage

(II) - ↑ Proinflammatory state $\rightarrow$ ↑ Blood-brain barrier permeability

- ↑ Cerebral cytokines (e.g. TNF, IL1B) leads to HE

Others: ↑ Cerebral lactate, manganese accumulation in basal ganglia, GABA receptor upregulation

Hepatic Encephalopathy

Grading and clinical features

Grade	Consciousness and Intellect	Clinical features	EEG
0	Normal	Nil	Normal
Subclinical (Minimal)	Normal	Nil except psychomotor testing	Normal
1	Inverted sleep pattern, forgetfulness, irritability	Tremor, apraxia, impaired handwriting	Slow 5 cps triphasic waves

Hepatic Encephalopathy

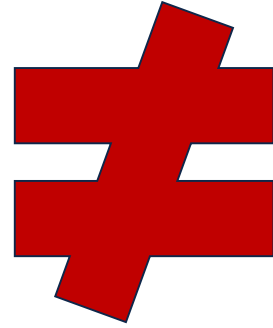
Grading and clinical features

Grade	Consciousness and Intellect	Clinical features	EEG
2	Lethargic, amnesia, ↓ inhibition	Flapping tremor, dysarthria, ↓ reflexes	Slow 5 cps triphasic waves
3	Somnolent but rousable, aggressive	Flapping tremor, ↑ reflexes, Babinski +	Slow 5 cps triphasic waves
4	Coma	Decerebrate	Slow 2-3 cps delta waves

Other Clinical Features:

- (Fetor hepaticus)
- Cerebral oedema
 - Uncommon in chronic HE but frequent in ALF

**Confusion
in
Cirrhosis**



**Hepatic
Encephalopathy**

Differentials:

Drug- or alcohol-related

Withdrawal state (e.g. delirium tremens)

Head injuries

CNS infections

Systemic infections

Metabolic disturbances (e.g. glucose)

Psychiatric

.....and more...

Diagnosis of HE

- ◆ Exclude other differentials

- ◆ Arterial ammonia

- normally 40% by bacterial action, 60% directly from protein breakdown

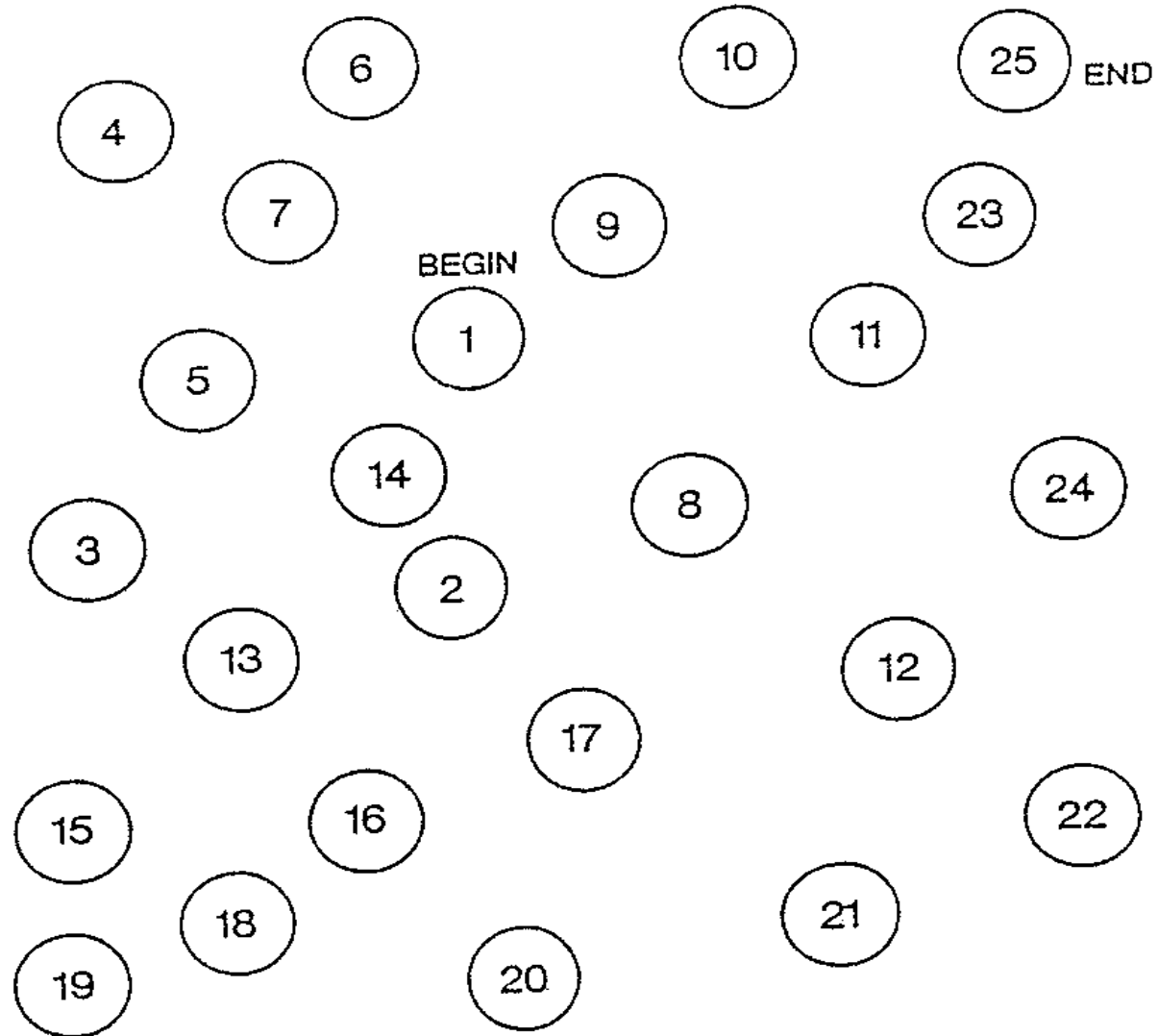
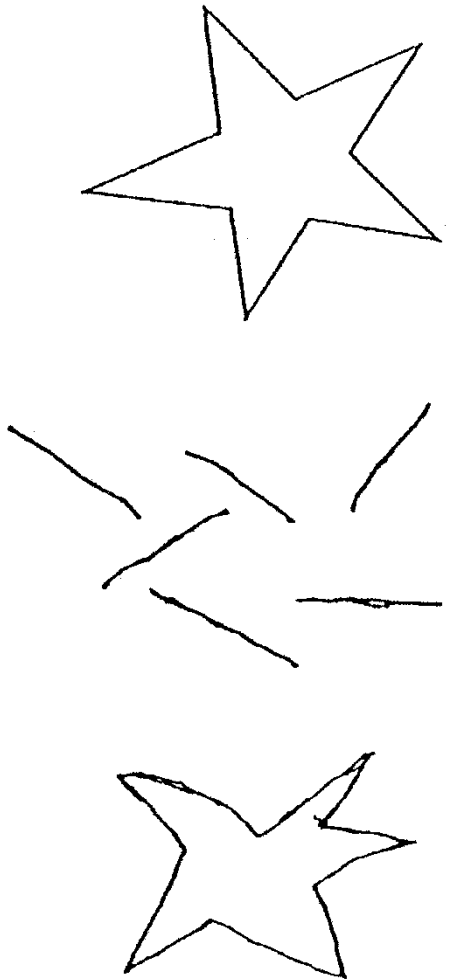
- not always raised; may not correlate with severity of HE

- ◆ EEG abnormalities

- ◆ Other features:

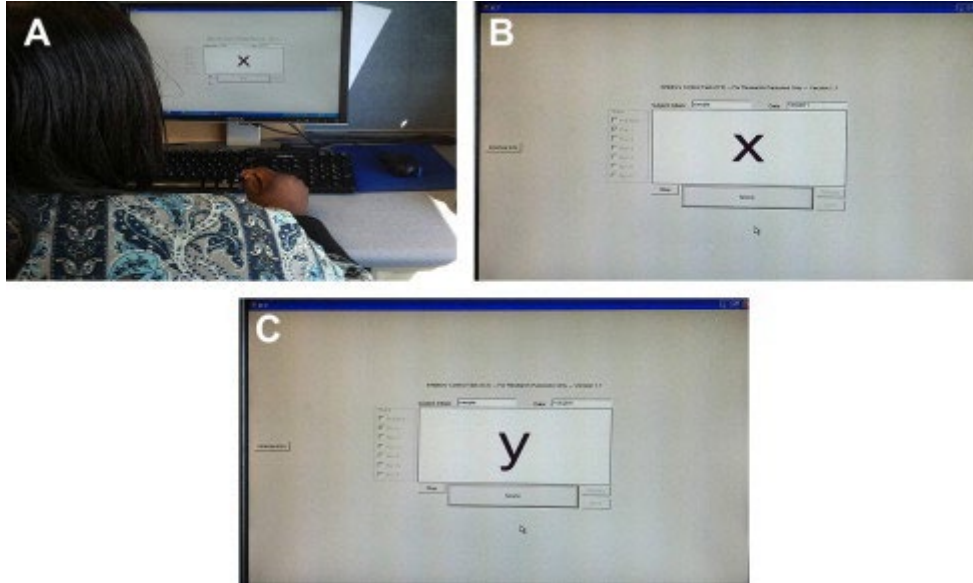
- fetor hepaticus
- flapping tremor
- psychometric tests, e.g., constructional apraxia (drawing a five-pointed star), number connection (Reitan's) test

Psychometric Tests



Complex Psychometric Tests for “Minimal” HE

And a smartphone app



Inhibitory Control Test



Time-
consuming....

Critical Flicker Frequency



Management of HE

- ◆ Identify precipitating factors and correct if correctable e.g., sepsis
especially look for SBP, ↓ K, withdraw sedatives
- ◆ Medical treatment:
 - ◆ Lactulose
 - ◆ Rifaximin
 - ◆ ?? Fecal microbial transplantation

Lactulose

- Nonabsorbable disaccharide
- Degraded in colon by urease-negative lactobacilli
- Mechanism of action:
 - Osmotic laxative
 - ↓ colonic pH
 - Inhibit urease-producing bacteria esp. E coli
 - Trap luminal $\text{NH}_3 \rightarrow \text{NH}_4$
 - draws NH_3 from mucosal blood to gut

2-3 times x soft stool / day

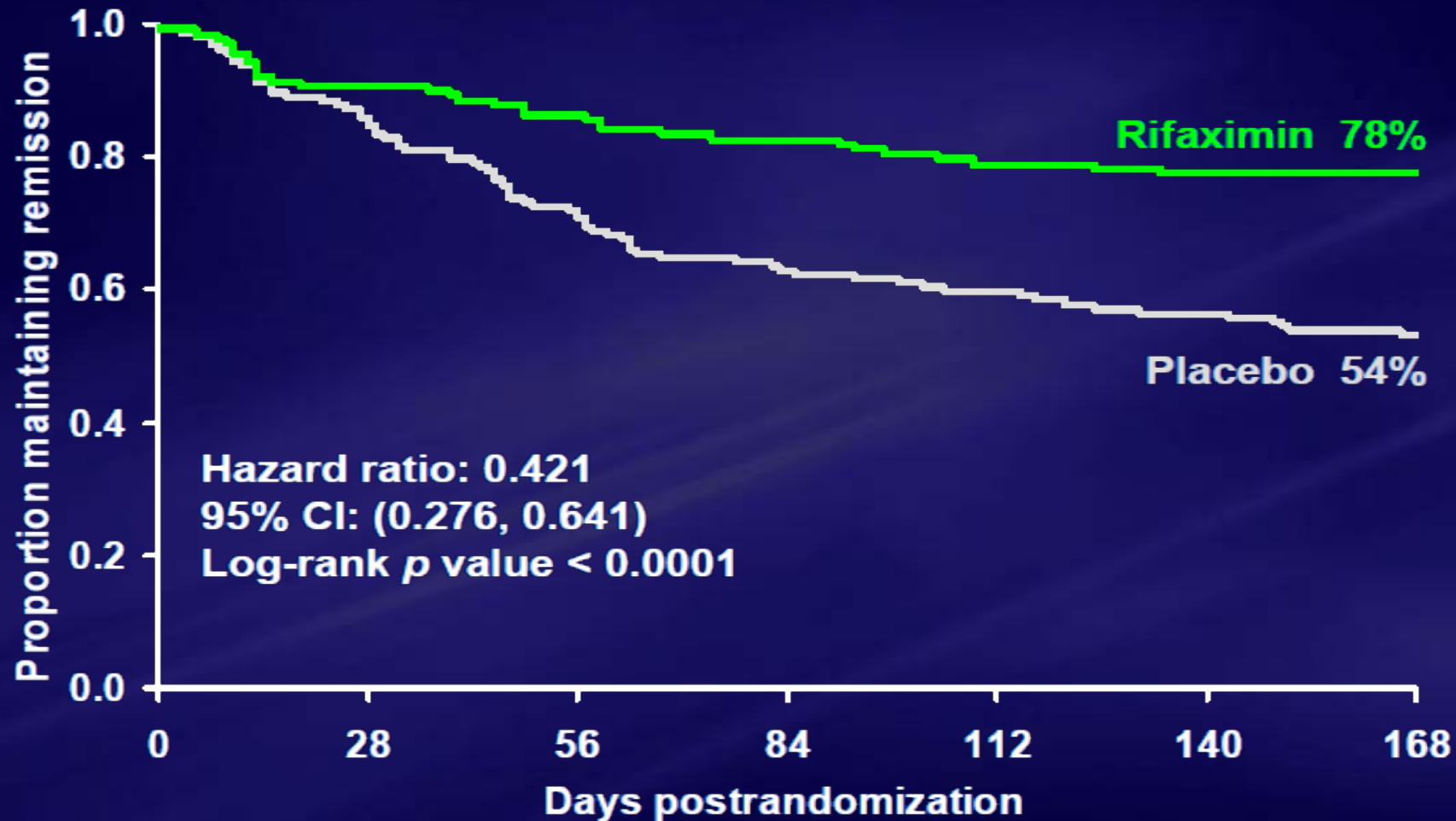
Can give enema form

Rifaximin

- Non-absorbable antibiotic
- Alteration of colonic flora (espc urease-producing)
- Combined used with lactulose effective
- Also effective in reducing recurrence of HE

- Neomycin: potential nephrotoxicity / ototoxicity, no longer recommended

Rifaximin significantly reduced the risk of HE breakthrough by 58% during the 6-month treatment



Fecal Microbiota Transplantation?

Time with tell...



HEPATOLOGY

HEPATOLOGY, VOL. 70, NO. 5, 2019



LIVER FAILURE/CIRRHOSIS/PORTAL HYPERTENSION

Fecal Microbial Transplant Capsules Are Safe in Hepatic Encephalopathy: A Phase 1, Randomized, Placebo-Controlled Trial

Jasmohan S. Bajaj ^{1,2}, Nita H. Salzman,³ Chathur Acharya,^{1,2} Richard K. Sterling,^{1,2} Melanie B. White,^{1,2} Edith A. Gavis,^{1,2} Andrew Fagan,^{1,2} Michael Hayward,³ Mary L. Holtz,³ Scott Matherly,^{1,2} Hannah Lee,^{1,2} Majdi Osman,⁴ Mohammad S. Siddiqui ⁴, Michael Fuchs,³ Puneet Puri,³ Masoumeh Sikaroodi,⁵ and Patrick M. Gillevet⁵

Management of HE

◆ Diet

- Avoid dietary protein restriction

(except in active GI bleeding)

- Recommended protein intake:
40-60g/day
- Dietitian input important

Increasing ammonia removal (Mod effect only)

Branch-chained amino acids (BCAA)

- Detoxifies ammonia via the production of glutamine
- Also function as nutritional supplement
- Mainly in protein-intolerant patients

L-ornithine-L-aspartate (LOLA)

- Converts L-ornithine to glutamate in muscle
- Conversion uses ammonia to produce glutamine

Management of HE

- Increasing ammonia removal

- Branch-chained amino acids (BCAA)
 - Detoxifies ammonia via the production of glutamine
 - Also function as nutritional supplement
 - Mainly in protein-intolerant patients
- L-ornithine-L-aspartate (LOLA)
 - Converts L-ornithine to glutamate in muscle
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Coagulopathy in Liver Failure

- Platelet dysfunction (quantitative & qualitative)
 - Platelet transfusion required if:
 - Bleeding or invasive procedure and <50
 - Prophylactically if <20
- Prolongation of PT and INR
 - PT/INR **important** prognostic indicators
 - FFP only if active bleeding/invasive procedure

Acute Liver Failure - Management

- **Supportive** (standard ICU care)
- Identifying and removing/**treating the insult** that caused the hepatic injury
- **Manage complications**
 - Infections
 - Hepatic encephalopathy
 - Increased ICP / Cerebral oedema
 - Renal failure
 - Metabolic complications: ↓Na, ↓glucose etc.
- High-volume plasma exchange*
- **Liver transplantation**



Eg. HBV → Treat HBV
Paracetamol → N-acetylcystine
Mushroom → Activated charcoal

Unproven / Experimental

- Liver support systems (e.g. MARS)
- N-acetylcystine for all etiologies
- G-CSF
- Plasmapheresis (high volume)

WCS Outline: Revisit

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THANK YOU!

*Email questions to
wkseto@hku.hk*

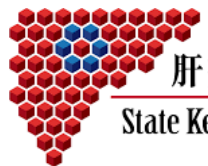
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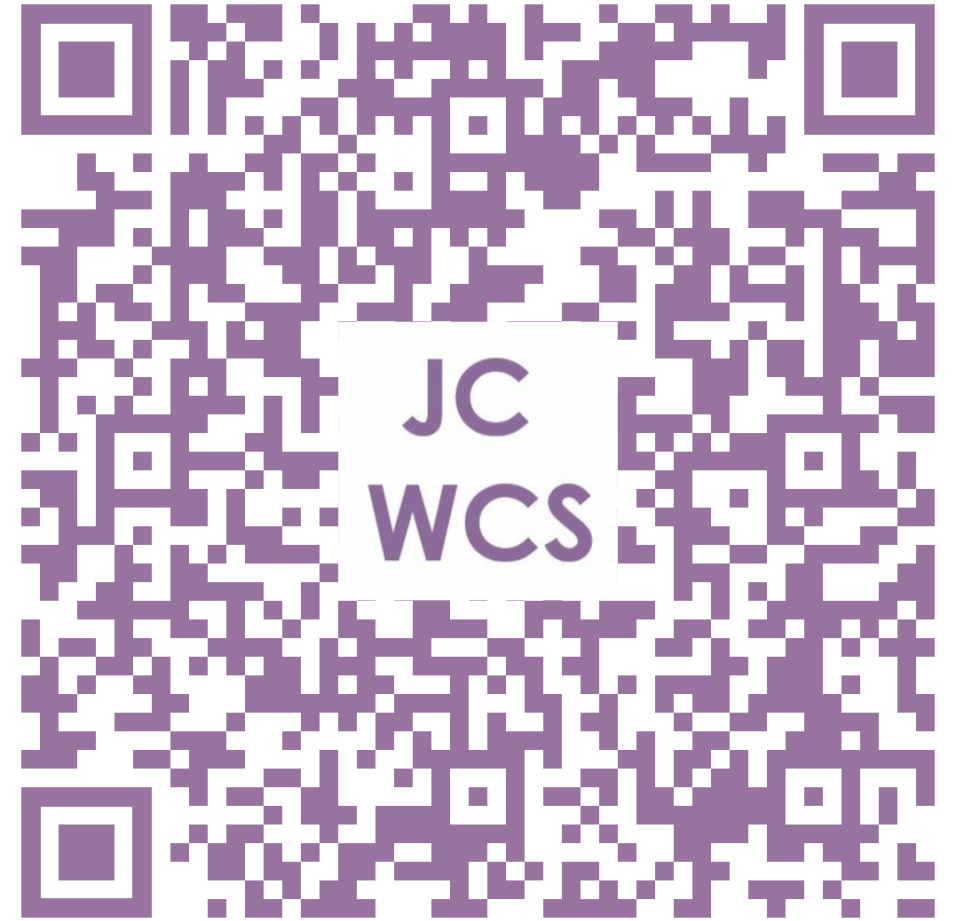
TELL US WHAT

JC Whole Class Session

YOU THINK

THANK

YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)